# Predicting NBA Offensive Rating: The Impact of Three-Point Shooting Efficiency and Volume

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# Abstract

This project aimed to predict NBA teams' offensive ratings using machine learning models based on their 3-point shooting statistics from 1996 onwards (the earliest year that ORTG data is available). We found that the multiple linear regression model, using 3-point attempt rate and 3-point percentage, proved to be the best-performing model, being able to explain approximately 67% of the variation in offensive rating. We found that shooting efficiency seemed to have a much greater impact on offensive rating than shooting volume (2.7x more), with a 1% increase in 3-point percentage resulting in +0.88 ORTG points, while a 1% increase in 3-point attempt rate resulted in only +0.33 ORTG points. We also implemented a polynomial regression model, which achieved a worse performance than the multiple linear regression model (only explaining around 56% of the variation in offensive rating), confirming that 3-point statistics and offensive rating have a mainly linear relationship. Analysis of residual plots across all models implemented showed reasonably met regression assumptions, with very small violations of the normality assumption. Overall, we found that 3-point metrics can be considered strong predictors of an NBA team's offensive success and that the importance of 3-point efficiency outweighs the importance of 3-point volume in modern NBA basketball. Finally, the project looked to develop metrics to accurately determine how impactful a 3-point shooter an individual player is, which is determined by both their own abilities and their impact on their teammates. This approach could help teams decide which players will best improve their three-point abilities. This connects to our main point, as we look to understand how this change in 3-point abilities connects to offensive efficiency.


# Introduction

The evolution of NBA strategy over the past few decades has been heavily influenced by the introduction of the 3-point shot and its rise in usage in offensive systems, with NBA teams often looking to generate as many open 3-point shots as possible in their offensive game plans. Since its introduction in 1979, teams have started to rely on the 3-point shot more and more, transforming how the game of basketball is played at the professional level. By analyzing NBA team statistics since 1979, we would be able to discover how much 3-point shooting actually impacts the success of a team and whether NBA teams should continue to revolve their game plans around it. This analysis focuses on the relationship between different 3-point shooting metrics (attempt rate, efficiency, etc.) and different success metrics of an NBA team (such as offensive rating and win percentage), trying to determine how correlated these relationships are, how they have evolved, and which shooting strategies are able to most effectively predict the success of an NBA offense. Additionally, we also aim to build machine learning models that can predict the offensive rating of an NBA team, using 3-point shooting metrics, which can be used to assist NBA teams in their offensive game planning. The key questions are as follows:

1. How does 3-point shooting impact the offensive efficiency of an NBA team?
2. How much do 3-point shooting rates affect winning rates of an NBA team?
3. Can we accurately predict offensive rating based on 3-point shooting metrics?

# Data Description

Summary of the Data Processing Pipeline:
1. Acquire NBA statistics data from the nba_api library
2. Clean data and convert to tables for visualization and analysis
3. Visualize using plotting libraries such as Plotly, Seaborn, and Matplotlib
4. Implement machine learning models and perform regression analysis

First, we acquired the NBA stats dataset by utilizing nba_api, an API client package that allows us to easily retrieve statistics from the official [NBA.com](NBA.com) website. We used the dataset to get yearly team statistics from 1979 to 2024. We then cleaned the raw data and converted them into tables within the Jupyter Notebook. The cleaning process consisted of converting the raw data into a Pandas DataFrame, as well as removing unnecessary columns. Since offensive rating data only became available after 1996, we made a separate DataFrame to retrieve advanced statistics (which included offensive rating). We then merged this DataFrame with the original basic statistics DataFrame using TEAM_ID as the identifier, creating one final dataset. To address missing values, we checked and replaced them, using NaN as a placeholder.

Next, after we cleaned our dataset, we added columns necessary to improve our analysis, including 3-point attempt rate, effective field goal percentage, and 3-pointers per 100 possessions. We did this to prepare our data for the machine learning models by choosing relevant features for predictive offensive rating modeling. Further analysis then included generating basic statistics and plots to visualize the relationships between 3-point metrics and offensive performance. We developed time series plots to visualize how 3-point shooting rates have increased over the years, correlation plots to show the relationship between 3-point shooting metrics and offensive rating, and heat maps to visualize where on the court previous NBA MVPs tended to attempt shots.

| | TEAM_ID | TEAM_NAME | GP | W | L | W_PCT | MIN | FGM | FGA | FG_PCT | ... | NET_RATING | PACE | SEASON | FG3A_RATE | EFG_PCT | YEAR | FG3A_PER_100 | FG2_PCT | FT_RATE | FG3A_PER_100_X_FG3_PCT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1610612737 | Atlanta Hawks | 82 | 56 | 26 | 0.683 | 3961.000000 | 2812 | 6307 | 0.446 | ... | 6.2 | 88.54 | 1996-97 | 0.290629 | 0.498097 | 1996 | 25.246960 | 0.481225 | 0.309973 | 9.088906 |
| 1 | 1610612738 | Boston Celtics | 82 | 15 | 67 | 0.183 | 3981.000000 | 3066 | 6967 | 0.440 | ... | -7.2 | 96.78 | 1996-97 | 0.191043 | 0.473590 | 1996 | 16.771758 | 0.461143 | 0.315631 | 5.886887 |
| 2 | 1610612766 | Charlotte Hornets | 82 | 54 | 28 | 0.659 | 3960.833333 | 2988 | 6342 | 0.471 | ... | 2.1 | 90.04 | 1996-97 | 0.217912 | 0.517739 | 1996 | 18.717968 | 0.483266 | 0.312835 | 8.011290 |
| 3 | 1610612741 | Chicago Bulls | 82 | 69 | 13 | 0.841 | 3945.833333 | 3277 | 6923 | 0.473 | ... | 11.8 | 91.50 | 1996-97 | 0.202658 | 0.511122 | 1996 | 18.699187 | 0.498913 | 0.266936 | 6.974797 |
| 4 | 1610612739 | Cleveland Cavaliers | 82 | 42 | 40 | 0.512 | 3971.000000 | 2704 | 5972 | 0.453 | ... | 1.9 | 84.31 | 1996-97 | 0.215003 | 0.493218 | 1996 | 18.572573 | 0.473763 | 0.296885 | 6.983288 |
| 5 | 1610612742 | Dallas Mavericks | 82 | 24 | 58 | 0.293 | 3960.970000 | 2812 | 6451 | 0.436 | ... | -6.5 | 90.65 | 1996-97 | 0.203844 | 0.469152 | 1996 | 17.690662 | 0.463980 | 0.297318 | 5.767156 |
| 6 | 1610612743 | Denver Nuggets | 82 | 21 | 61 | 0.256 | 3986.000000 | 2934 | 6687 | 0.439 | ... | -6.6 | 93.67 | 1996-97 | 0.255870 | 0.486317 | 1996 | 22.275919 | 0.461817 | 0.297891 | 8.286642 |
| 7 | 1610612765 | Detroit Pistons | 82 | 54 | 28 | 0.659 | 3966.000000 | 2827 | 6095 | 0.464 | ... | 5.9 | 86.09 | 1996-97 | 0.245939 | 0.511567 | 1996 | 21.234159 | 0.488468 | 0.327317 | 8.238854 |
| 8 | 1610612744 | Golden State Warriors | 82 | 30 | 52 | 0.366 | 3980.681667 | 2996 | 6566 | 0.456 | ... | -5.0 | 93.58 | 1996-97 | 0.207585 | 0.492918 | 1996 | 17.762290 | 0.483375 | 0.332013 | 6.270088 |
| 9 | 1610612745 | Houston Rockets | 82 | 57 | 25 | 0.695 | 3970.908333 | 3037 | 6484 | 0.468 | ... | 4.7 | 92.73 | 1996-97 | 0.283621 | 0.520126 | 1996 | 24.185085 | 0.509365 | 0.307218 | 8.827556 |
| 10 | 1610612754 | Indiana Pacers | 82 | 39 | 43 | 0.476 | 3986.000000 | 2851 | 6254 | 0.456 | ... | 1.3 | 90.44 | 1996-97 | 0.180684 | 0.490246 | 1996 | 15.237160 | 0.472482 | 0.373521 | 5.805358 |
| 11 | 1610612746 | Los Angeles Clippers | 82 | 36 | 46 | 0.439 | 3986.000000 | 2989 | 6696 | 0.446 | ... | -2.8 | 92.65 | 1996-97 | 0.199970 | 0.481780 | 1996 | 17.624682 | 0.469479 | 0.309737 | 6.239138 |
| 12 | 1610612747 | Los Angeles Lakers | 82 | 56 | 26 | 0.683 | 3986.000000 | 3018 | 6642 | 0.454 | ... | 4.2 | 92.61 | 1996-97 | 0.225836 | 0.495860 | 1996 | 19.752384 | 0.479774 | 0.350798 | 7.249125 |
| 13 | 1610612748 | Miami Heat | 82 | 61 | 21 | 0.744 | 3961.000000 | 2822 | 6235 | 0.453 | ... | 6.0 | 89.54 | 1996-97 | 0.299118 | 0.506977 | 1996 | 25.400829 | 0.490618 | 0.324298 | 9.245902 |
| 14 | 1610612749 | Milwaukee Bucks | 82 | 33 | 49 | 0.402 | 3946.000000 | 2967 | 6303 | 0.471 | ... | -2.0 | 90.85 | 1996-97 | 0.145962 | 0.496430 | 1996 | 12.349491 | 0.490990 | 0.333809 | 4.347021 |
| 15 | 1610612750 | Minnesota Timberwolves | 82 | 40 | 42 | 0.488 | 3956.000000 | 2937 | 6436 | 0.456 | ... | -1.7 | 92.04 | 1996-97 | 0.169826 | 0.485162 | 1996 | 14.482039 | 0.480255 | 0.338720 | 4.909411 |

# Methods

In regard to addressing the research question of "How does 3-point shooting impact the offensive efficiency of an NBA team?", we selected different metrics in order to capture the volume and the efficiency of the shot in close regard to the nature of the sport. First, we established the historical evolution of the shot through Phase 2 visualizations, which included MVP shooting charts and overall league 3-point tendencies before and after Stephen Curry's league-wide influence. Once completing this, we moved into quantifying the impact of the 3-pointer on modern offenses, analyzing the direct impact on its performance using multiple regression models. Offensive rating was selected as our dependent variable because it measures points scored per 100 possessions, which is crucial because it helps to normalize other factors like pace of play and game time and, overall, tries to diminish the direct impact of playstyles, which allows us to focus directly on the 3-point shot's impact. Discussions of using raw points per game or percentage of points per game scored with 3-pointers led us to understanding that offensive rating was the perfect dependent variable. It captures overall offensive efficiency while also accounting for the number of opportunities, diminishing the impact of offensive strategy and getting closer to directly answering whether three-point shooting makes offenses truly better.

We developed several regression models to determine and analyze the 3-pointers' impact from different angles and perspectives. We selected linear regression models for our analysis because they enabled us to use a variety of metrics and quantify their relationship with our target (offensive rating) using coefficients and visuals. Furthermore, the high volume and stability of basketball metrics are well suited to the robust nature of linear models. Our first model followed the equation $OFF_RATING = \beta_0 + \beta_1(FG3A_PER_100) + \beta_2(FG3_PCT) + \beta_3(FG3A_PER_100 \times FG3_PCT) + \varepsilon$. In this equation FG3A_PER_100 represents the number of three-point attempts per 100 possessions, and FG3_PCT represents the three-point shooting percentage. The key detail of this model was using an interaction term ($FG3A_PER_100 \times FG3_PCT$), which multiplies volume and efficiency together, which in turn tests if the benefit of shooting more threes depends on how accurately you shoot them, or if it's just dependent on volume. Our second model uses polynomial regression, which attempts to find non-linear relationships in the same topic. It uses the formula $OFF_RATING = \beta_0 + \beta_1(FG3A_RATE) + \beta_2(FG3A_RATE^2) + \beta_3(FG3A_RATE^3) + \varepsilon$, where FG3A_RATE measures the proportion of shot attempts that are behind the three-point line. This model can determine if there is an optimal shooting rate or if offenses benefit from chucking up threes without a limit. After finding that the polynomial regression wasn't effective and finding success in combining both volume and efficiency, we moved to our third model serving as a baseline with the equation $OFF_RATING = \beta_0 + \beta_1(FG3A_RATE) + \beta_2(FG3_PCT) + \varepsilon$, assuming that both volume and efficiency have an independent effect on offensive rating. This framework provides interpretations where $\beta_1$ represents the effect of shooting more shots while efficiency stays constant, and $\beta_2$ represents the effect of higher efficiency on the same volume.

Finally, we created metrics to label how good each NBA player is at shooting and how much they impact their teammates' shooting, labeled "shooting score" and "impact score." The following are their equations:

- **Impact Score:** $\sqrt{(FG3M_ON)} * \quad (FG3_PCT_OFF \ - \ FG3_PCT_ON)$
    - FG3M: Team's total 3-pointers made when player X is playing (not including his makes)
    - FG3_PCT_OFF: Team's 3-pointer percentage when player X is not playing
    - FG3_PCT_ON: Team's 3-pointer percentage when player X is playing (not including

his shots/attempts)

- **Player score:** $FG3_PCT \ * \ e^{(FG3_PCT - AVG_FG3PCT)} \ * \ log(1 + FG3M)$
    - FG3_PCT: Player's 3-Point Percentage
    - AVG_FG3PT: League Average 3-Point Percentage
    - FG3M: Player's Total 3-Pointers Made

Our equations underwent several iterations until we chose these formulas. We compared them against our database of 75 NBA players who changed teams in the last 5 years to evaluate their impact. Collecting this data was difficult because we could not rely on existing APIs and had to track individual occurrences.

**Highest performers:**

- **Player Score: CJ McCollum, Mike Conley, Malik Beasley**
- **Impact Score: Kyrie Irving, Donovan Mitchell, Pascal Siakam**

**Worst performers:**

- **Player Score: Jarrett Allen, Rudy Gobert, Jakob Poeltl**
- **Impact Score: Kelly Olynk, Domantas Sabonis, Jalen McDaniels**

These results make sense given the abilities of these players. We wanted to see whether these simple metrics could be combined into a single statistic that would allow us to score each team's ability as the sum of its players' scores, and use that to predict each player's overall impact and how they affect their team's 3-point percentage and attempts. Since these metrics returned such a different range of values, we standardized them. Furthermore, including the team's baseline 3-point shooting performance provided important context for the model, allowing it to understand each team's existing capabilities when a new player was acquired."

All our models were evaluated using $R^2$ in order to measure the explanatory power, mean squared error to assess prediction accuracy, and cross-validation methods to detect instances of overfitting. We also examined residual diagnostic plots to verify our statistical assumptions, like constant variance and distribution errors. Using multiple complementary models allowed us to determine robust conclusions while testing various theoretical assumptions and trying to limit the limitation of a singular model. The multiple-model approach provides extensive evidence for answering the question of whether and how 3-point shooting impacts an NBA team's offensive performance in today's NBA.

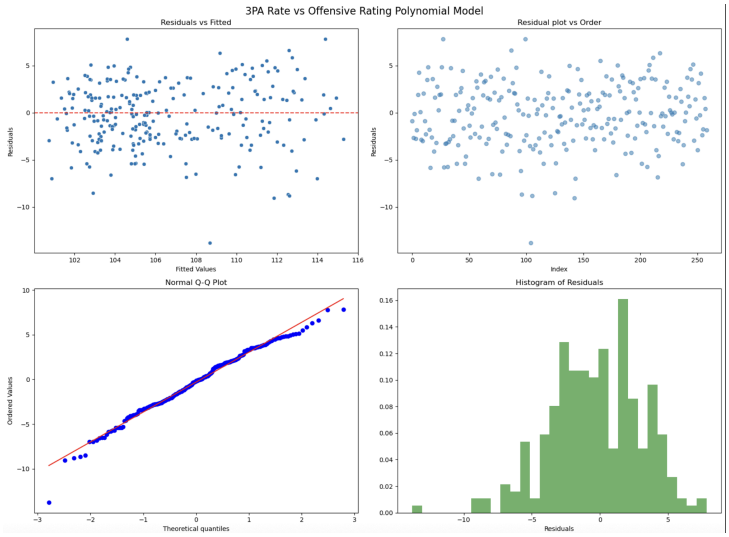
# Results

First, we analyzed Model 1 utilizing a multiple linear regression model with the aforementioned interaction term. This model achieved a test MSE of 7.773, an RMSE of 2.788, and an $R^2$ value of 0.654. All of these metrics are comparable to the training results, which resulted in a training MSE and $R^2$ of 8.329 (RMSE: 2.886) and 0.681, respectively. This indicates a minimal amount of overfitting with only a 2.7% difference between train and test $R^2$. The model uses three features: three-point attempts per 100 possessions (FG3A_PER_100), three-pointer shooting percentage (FG3_PCT), and the interaction term (FG3A_PER_100 x FG3_PCT). The fitted equation came out to OFF_RATING = 97.49 - 1.20(FG3A_PER_100) + 0.36(FG3_PCT) + 4.42(FG3A_PER_100 × FG3_PCT). In regard to the plots, it is evident that the model satisfies our regression assumption in a reasonable manner. In the plot analyzing the actual vs. predicted data, there is a strong alignment with the perfect prediction line, with clusters around this line following quite

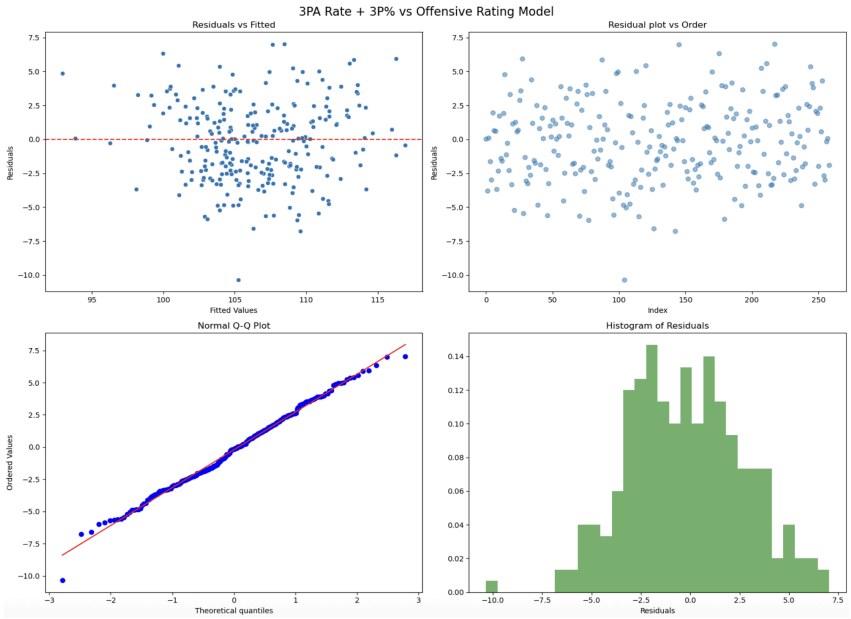
tightly, displaying a clear linear relationship. The residual plot has a dispersion around 0, with most clusters forming near 0 with a small amount of heavy outliers, meeting the linear assumptions in a satisfying way. The histogram shows a reasonably symmetric, bell-shaped curve centered somewhat near zero, indicating that the assumption is normal; however, there are some parts of the histogram that don't follow the normality pattern. The Feature Importance plot is the most interesting here; it reveals that the interaction term (FG3A_PER_100 × FG3_PCT) has the largest positive coefficient over the individual predictors, highlighting the importance of both volume and efficiency. Interestingly to note, there exists a negative coefficient of -1.20 for FG3A_PER_100, directly showing that volume alone does not improve offense, but efficiency alone (FG3_PCT) does, as it had a coefficient of 0.4. This goes to show that both shooting more threes and efficiency have improved offensive rating together; however, when you shoot more at the same effectiveness or shoot the same more effectively, the changes to offensive rating are minimal.

Secondly, we analyzed Model 2 using a polynomial regression model to try and find non-linear relationships. To determine the best polynomial degree to use for our model, we tested degrees ranging from 1 to 5 and determined that degree 3 was the most optimal, achieving a test Mean Squared Error (MSE) of 11.273 and an $R^2$ value of 0.558, which was comparable to its train MSE and $R^2$, being 11.813 and 0.5222, respectively, indicating minimal overfitting. When analyzing the residual plots, we see that the model reasonably satisfies our regression assumptions. We can see that the Residuals vs Fitted plot shows a random scatter around zero with no apparent pattern, indicating that the linearity and homoscedasticity assumptions are met. The Residuals vs. Order plot shows a uniform distribution across the indices, showing that the independence assumption is satisfied. The Q-Q plot shows strong alignment with the diagonal line, with very minor deviations in the tails. However, these deviations are normal in real-world datasets, so it is not a sign of assumption violation. The histogram of residuals shows a slight left-skewedness; however, it is quite minimal, and the overall shape still remains reasonably symmetric, indicating that the normality assumption is reasonably met. While testing multiple polynomial degrees, we found that adding more degrees resulted in a negligible benefit. Degree 1 had an $R^2$ value of 0.557, which is a tiny difference. This showed us that the

relationship between offensive rating and 3-point attempt rate is mainly linear, rather than curved.



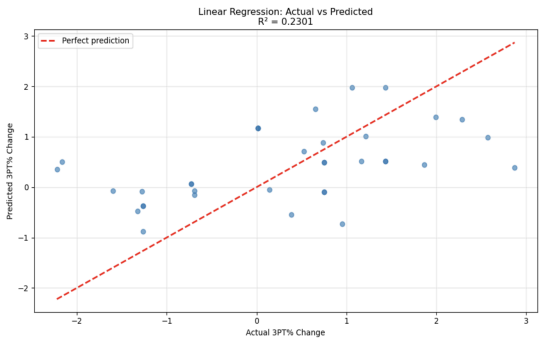

Moreover, we analyzed Model 3 using another multilinear regression model, as it was more effective than polynomial. After finding success combining both efficiency and volume, we decided to use 3-point attempt rate and 3-point shooting percentage as our predictors. This model achieved an $R^2$ value of 0.665, our highest so far, and an MSE of 8.543. When looking at the residual plots, we can see it has similar results to Model 2. The Residuals vs Fitted plot shows a random scatter around 0, which means that the linearity and homoscedasticity assumptions are met. The Residual vs Order plot also shows uniform distribution of points across the indices, which means that the independence assumption is satisfied. The Q-Q plot shows strong alignment with the diagonal line with slight deviations in the tails, similar to Model 2. The histogram also shows a slight left-skewedness but still maintains a reasonably bell-shaped and symmetric shape, indicating that the normality assumption is reasonably met.

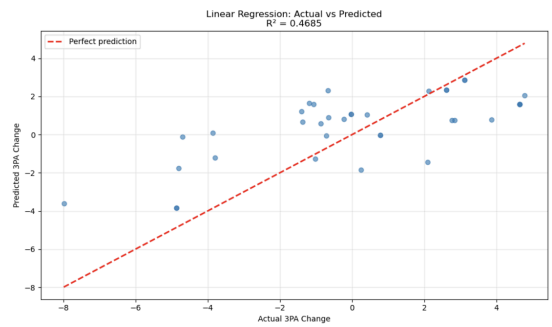


Lastly, we used linear regression for model 4, using our defined metrics as features, along with a team's 3-point percentage and attempts before acquiring a player, and the player's minutes on the new team. The strength of our model increased when we used our custom metrics rather than creating additional features, which consisted of the individual factors of those metrics. A team's current capabilities are obviously important, as we need to know the environment the player will work with. Finally, we concluded that minutes per game after joining a team is an acceptable feature, even though it is not predetermined, as the new squad entirely controls this statistic. Our predictor variables were the change in 3-point percentage and attempts, as they provide a complete picture of changes in shooting patterns. We benchmarked our linear regression against a polynomial regression model, which it outperformed. Below are our results:

3-Point Percentage:
$R^2 = 0.2301$, MSE = 1.147

3-Point Attempts:
$R^2 = 0.4685$, MSE = 2.0880




The results were interesting. We wanted to maintain colinearity, so we simply plugged in our features into the linear regression model. In a later iteration, we found that multiplying existing features together to create new features, such as impact_score and 3 PT attempts before trade, yielded higher $R^2$ scores. Nevertheless, predicting a player's direct impact on a team remains complex due to a variety of factors.

# Discussion

These findings demonstrate that there is a relationship between three-point shooting and more production on NBA offenses in the modern game. The linear regression models we used provided relatively strong results, with our highest correlation coefficient being from Model 3, where we determined that a combination variable of both volume and efficiency led to the best results. The linear regression models all proved to be effective and tested different assumptions about a wide variety of aspects surrounding three-point shooting's direct effect on offensive rating, while doing our best to limit themes such as pace of play and amount of opportunities. The models' $R^2$ values ranging from 0.558 to 0.665 are particularly encouraging; even though it isn't the strongest correlation from a directly mathematical perspective, the reality is that with real-world data and with a sport with so many variables like basketball, the $R^2$ values are strong.

With all of the success in this regard, there are many areas of improvement that could happen within our strategy. Firstly, the implementation of a lot of models allowed us to expand into different possibilities relating to the research question; however, it led us to a limited exploration of each variable. The deeper exploration of each variable would've cemented our models as stronger and possibly be able to limit other

factors more. Additionally, even with our efforts to try and isolate our dependent variable, the reality is that there are several other factors where we had no control that could have skewed our data. The reality of the game with different player profiles, coaching styles, offensive strategies, and defensive opponents are factors that do interfere with our conclusions. With that being said, the model still proves that there is a strong correlation between shooting more and more effectively from behind the three-point line and having a direct effect on offensive rating as a league-wide trend; whether it is applicable on a game-to-game or team-to-team basis has to be further examined.

Relatively lower $R^2$ scores for predicting the impact of players on 3-point shooting can be attributed to two main limitations. For starters, it was a priority to collect data only from the last few years for player transactions, as the game is consistently evolving in many ways, and even data from 10 years ago would be less relevant. We also focused on transactions in which the player's new team had a minimum number of other changes, so we could better understand the player's direct impact. Nevertheless, obtaining only 75 data points was certainly a limitation, and in the future, we can learn to use older data to highlight the trends we seek. Furthermore, a goal of this project was to create metrics that give a scalar value of how effective NBA players are. These metrics do not factor in other changes in role, fit, and other factors that are hard to quantify and predict. In a more extensive future project, we can explore how to more accurately track a player's performance by quantifying these factors and using a more comprehensive neural network to make better predictions. Nevertheless, it's generally very difficult to predict exactly how a player would perform on another team, and regardless of advanced techniques, consistently making correct predictions would remain quite challenging.

**Future research direction** could include questions and themes such as

- What offensive strategy has the greatest success rate from the three-point line?
- Is it good offenses making the three-point shot more effective, or is it good three-point shooting making offenses more effective?
- An expansion of these models in an attempt to limit real-world basketball-related limitations in order to have a team-by-team or game-by-game impact.
- Analyzing the impact of good defense on these models.

An expansion of continuing to limit factors and understand further components surrounding the overall question of the impact the three-point shot has on offenses is crucial to determining this shot's impact on the game of basketball. The data proved that a correlation exists and that it is quite evident, so an expansion to these models would achieve stronger correlations and conclusions for real-life applications.